

Spatial Regression

HES 505 Fall 2025: Session 26

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Motivation

- Last class: **Moran's I** showed evidence of spatial autocorrelation
- Ordinary linear models assume **independent** residuals
- But in spatial data, nearby units often influence one another
- Today: What happens when we **don't** account for spatial structure, and what SAR/SEM models do instead

When LM Fails

Linear model (LM):

$$\begin{bmatrix} y \\ X \end{bmatrix} = \begin{bmatrix} \beta \\ X \end{bmatrix} + \begin{bmatrix} \varepsilon \\ 0 \end{bmatrix}, \quad \begin{bmatrix} \varepsilon \\ 0 \end{bmatrix} \sim N\left(0, \begin{bmatrix} \sigma^2 I & 0 \\ 0 & 0 \end{bmatrix}\right)$$

- Coefficient estimates remain **unbiased** (if X is exogenous)
- **Standard errors are wrong** $\rightarrow$ misleading p-values
- **Residuals remain autocorrelated** $\rightarrow$ violates assumptions

Two Types of Spatial Dependence

1. **Spatial Lag (SAR/LAG)** — outcomes spill over
2. **Spatial Error (SEM)** — unobserved spatial process

SAR (Spatial Lag) Model

$$y = \rho W y + X\beta + \varepsilon$$

- ρ : strength of spillover / interaction
- Direct effect: $X \rightarrow y$
- Indirect effect: neighbors' $X \rightarrow y$

SEM (Spatial Error) Model

$$\begin{aligned} &[y = X\beta + u, \quad u = \lambda W u + \varepsilon] \end{aligned}$$

- Spatial structure is in the **errors**, not y
- λ describes strength of spatial autocorrelation in the **unobserved** process
- Coefficients β retain their LM meaning

SAR vs SEM: How to Tell Them Apart?

- SAR: outcomes influence neighbors (spillover)
- SEM: residuals autocorrelated due to unobserved spatial process
- Use theory + residual diagnostics (Moran's I)

What Students Should Learn

- LM ignores spatial dependence → unreliable inference
- SAR: interactive spillover model → interpret effects, not raw ()
- SEM: correlated errors → () interpretation same as LM
- Both help resolve residual autocorrelation
- Model selection guided by theory + diagnostics

